

Retrieval-Augmented Generation

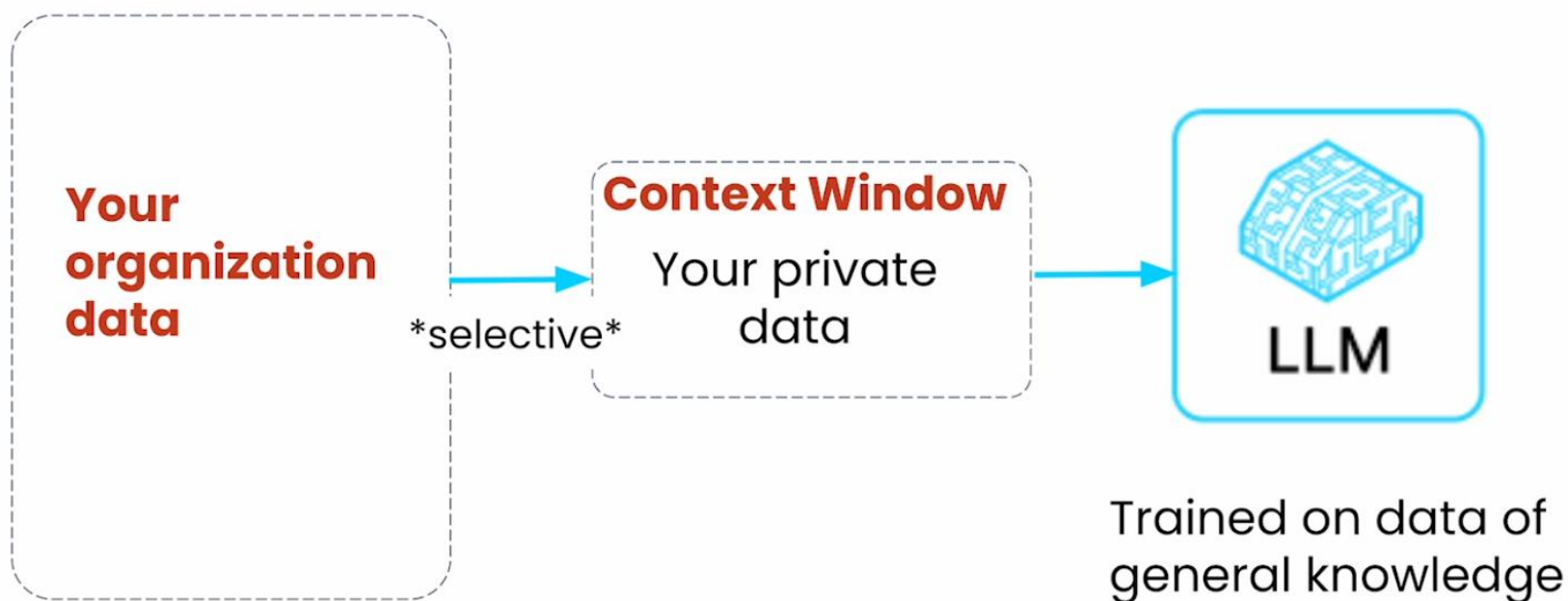
Hailiang Chen

HKU Business School

The University of Hong Kong

Limitations of LLMs

- Trained on general public data
- Frequent hallucinations
- Limited context window for private data

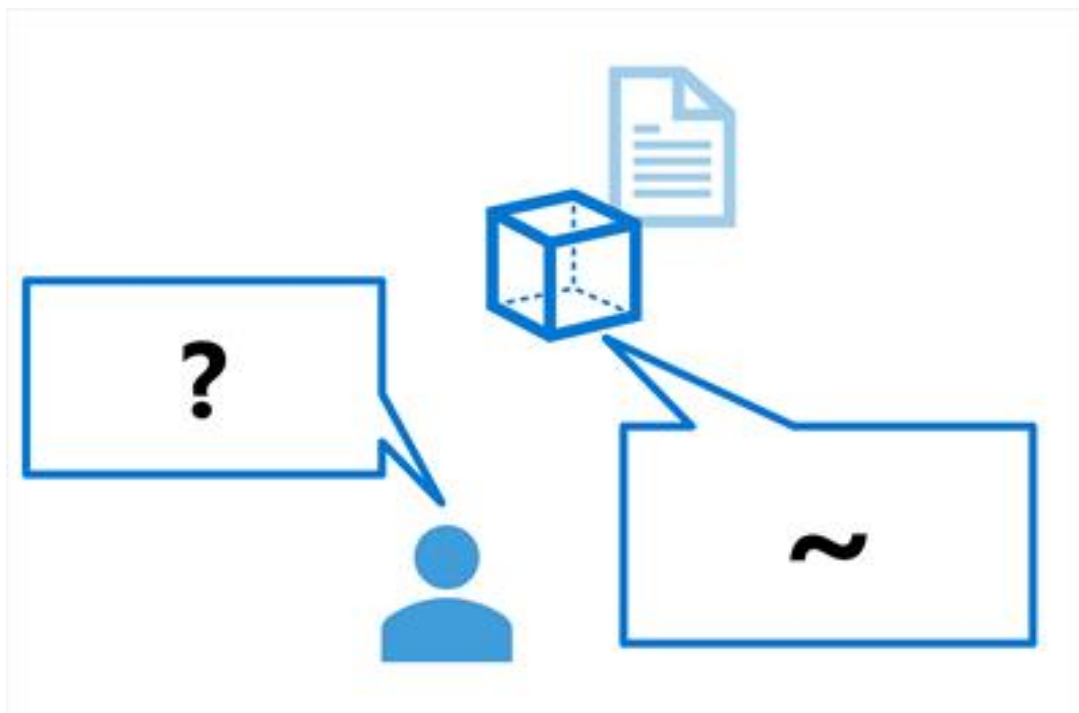


Why RAG?

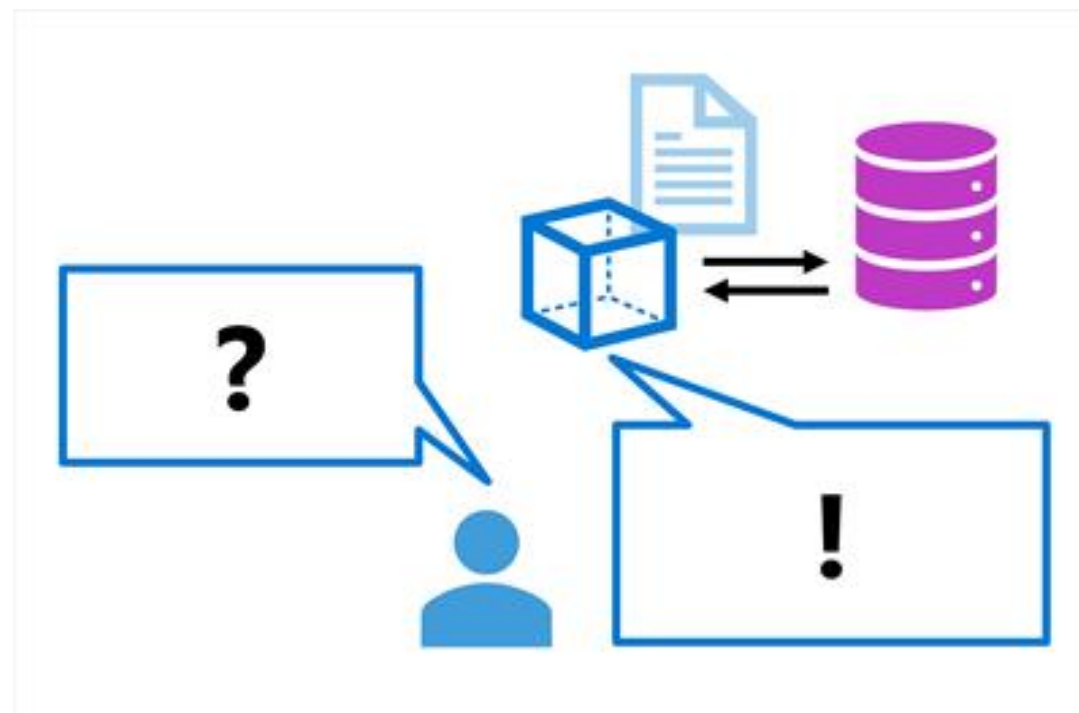
- Information richness: ensures text responses are **up to date** and **relevant**. It, therefore, enhances performance on domain specific tasks by accessing the internal knowledge base.
- Reduces fabrication or **hallucination** by utilizing verifiable data in the knowledge base to provide context to the user queries.
- It is **cost effective** as they are more economical compared to fine-tuning an LLM.

Grounding

Ungrounded



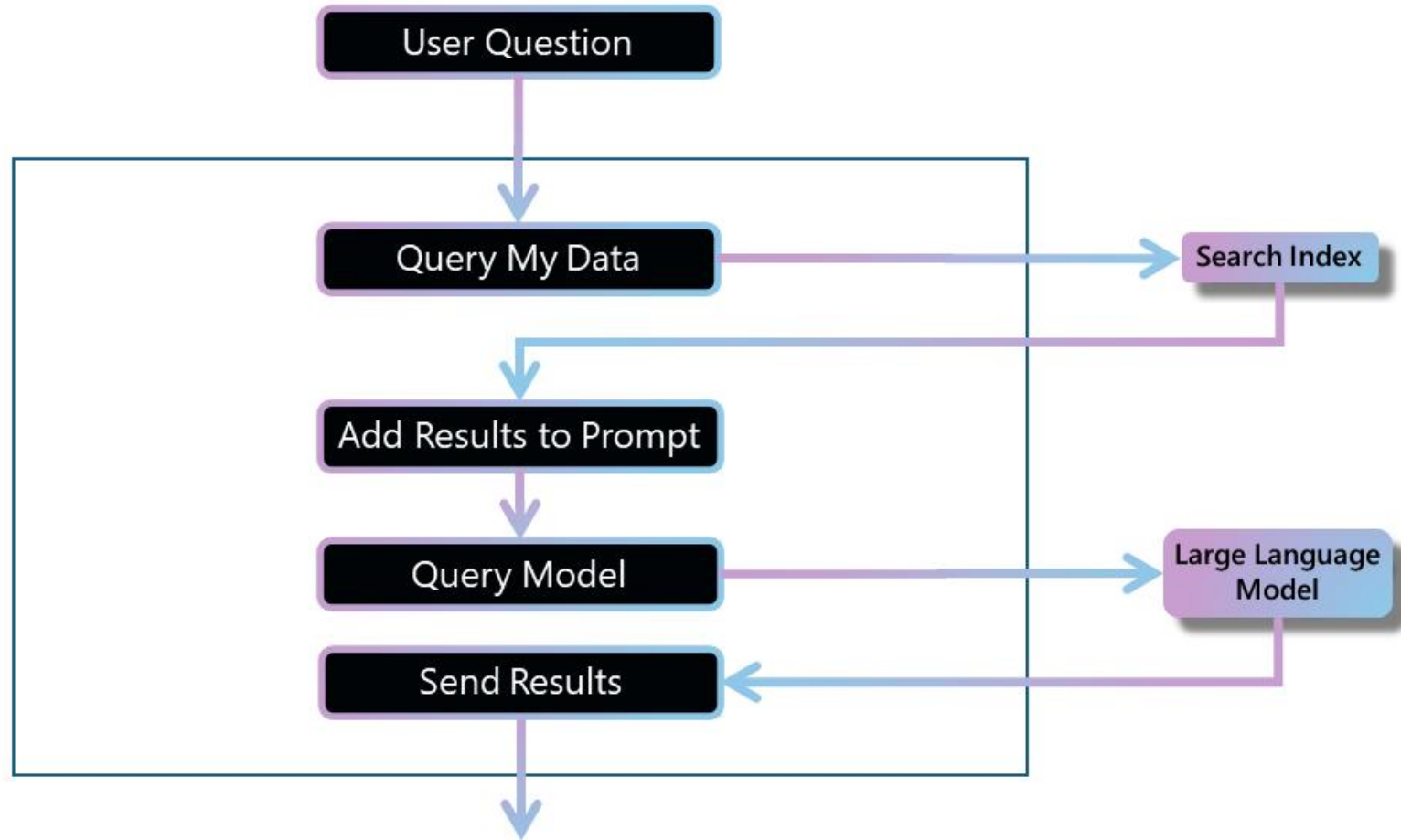
Grounded



Retrieval-Augmented Generation (RAG)

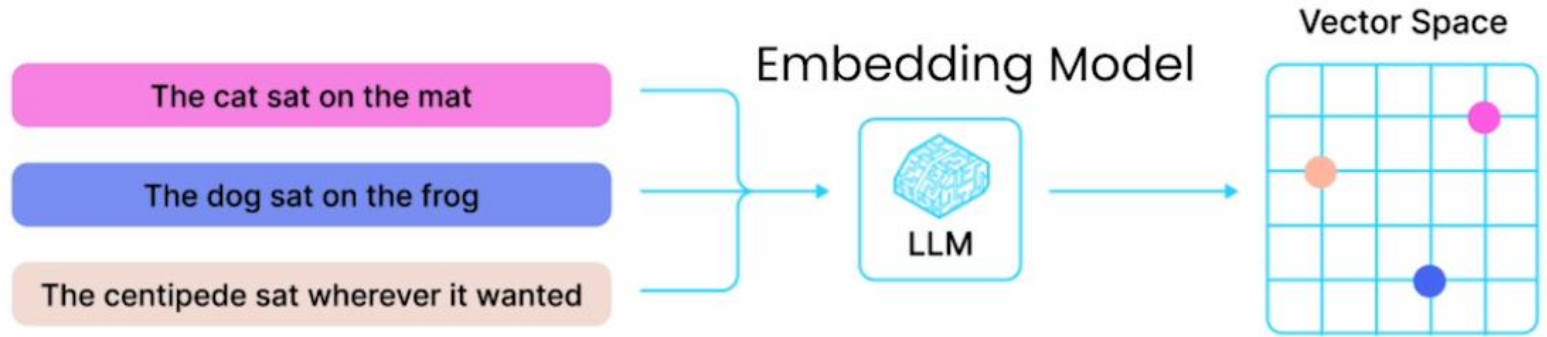
- Create Query: Generate a query based on the user's input.
- Perform Search: Use the query to search for relevant information in the knowledge base (e.g., database, Internet, vector store).
- Extract Information: Extract relevant content from the search results.
- Context Integration: Use the extracted information as context for generating an answer.
- Generate Answer: The large language model combines the context and user input to produce a response.

RAG - Retrieval Augmented Generation



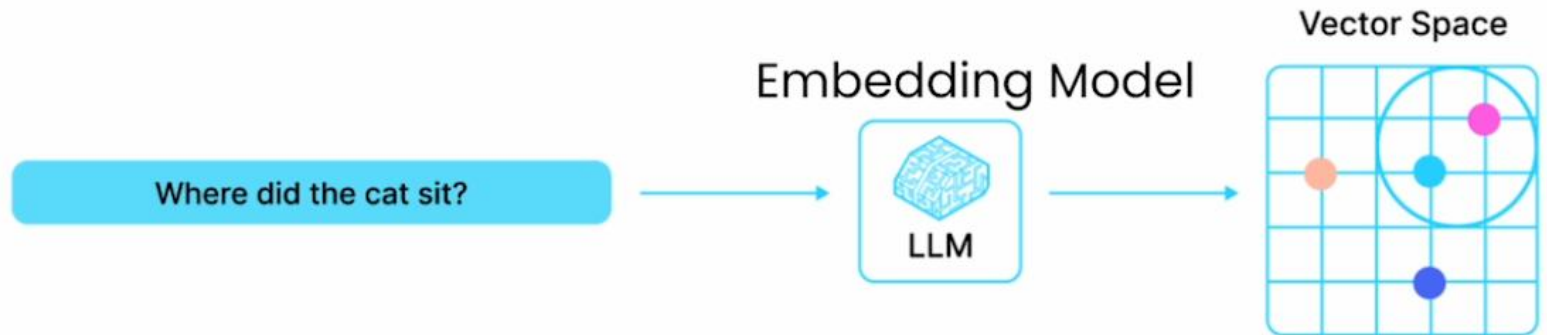
Retrieval Augmented Generation: RAG

1
Embed



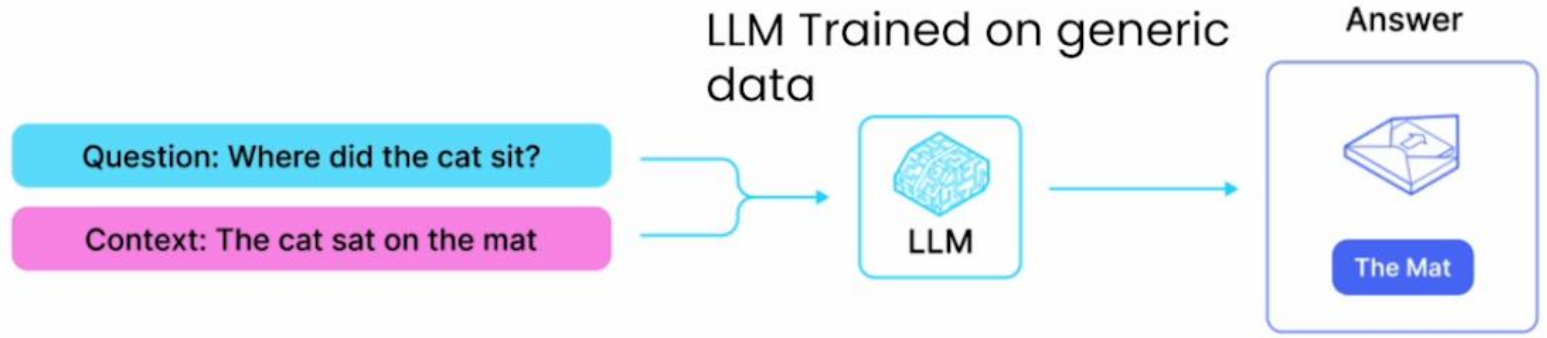
Retrieval:
find relevant
context

2
Retrieve

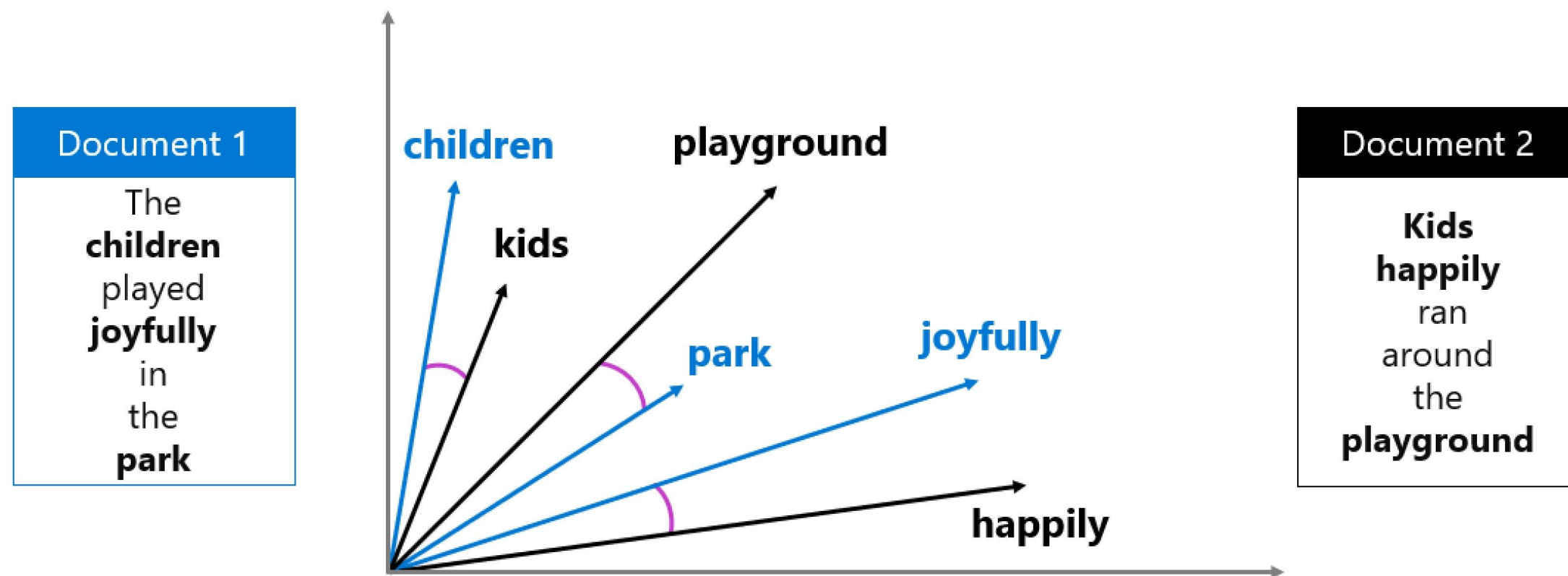


Generate:
answer the
question

3
Query



An embedding is a mathematical representation of text as a vector — a list of floating-point numbers that captures the meaning of words, sentences, or documents.



ASK THE POST AI



New chat



History



More

Good morning

Ask us about today's news.

Today's news

Tell me the latest update

Give me advice

Help me with

✦ What message is the "Best Friends Forever" statue intended to convey?

✦ What impact will Adelita Grijalva's win have on the House majority?

✦ What prompted the Log Cabin Republicans to allegedly heckle musician Yasmin Williams at the Kennedy Center?

✦ What is causing the surge in NDM-CRE bacteria infections in the U.S.?

What do you want to know?



Answers are AI generated from Washington Post reporting. Because AI can make mistakes, verify information by referencing provided sources for each answer.



The Washington Post

ASK	THE POST	AI
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What do you want to know?



Policy and Regulation Chatbot

- <http://47.113.119.241/> (demo site only; may go offline without notice)
- Report: <http://hlchen.com/doc/rag.pdf>
- Knowledge Base: Over 15K government documents, including laws, administrative regulations, recent policies, policy interpretations, and departmental public information.

19:36



您好，我是政策法规问答小助手。有什么可以帮您的吗？

人身保险产品的信息披露要求有哪些？

19:36



人身保险产品的信息披露要求包括以下几个方面：

1.
信息披露渠道：
1. 保险公司可以通过官方网站、官方公众服务号等自营平台披露产品信息。

2. 利用中国保险行业协会等行业公共信息披露渠道。

3. 通过保险公司授权或委托的合作机构和第三方媒体进行披露。

4. 在保险公司产品说明会等业务经营活动中披露。

5. 其他根据公司经营管理需要向保险消费者披露的渠道。
2.
信息一致性：

关于人身保险产品的信息披露渠道，是否有相关的监管政策对保险公司进行规范？

在人身保险产品的信息披露中，如何确保信息的准确性和完整性？保险公司有哪些责任和义务？

对于违反人身保险产品信

请输入...

发送

11. 产品信息变更：

1. 产品信息发生变更时，保险公司应在10个工作日内更新官方网站上的信息。

12. 信息披露管理：

1. 保险公司应建立产品信息披露内部管理办法，完善内部管理机制，强化销售过程与售后服务的信息披露监督管理。

13. 产品信息披露材料管理：

人身保险产品信息披露管理办法_国家金融监督管理总局__2022年11...30日起施行.pdf 片段：05c...

人身保险产品信息披露管理办法_国家金融监督管理总局__2022年11...30日起施行.pdf 片段：d29...

人身保险产品信息披露管理办法_国家金融监督管理总局__2022年11月...30日起施行.pdf 片段：f1d...

中国银保监会关于印发一年期以上人身保险产品信息披露规则的通知.pdf 片段：aa4...

中国银保监会关于印发一年期以上人身保险产品信息披露规则的通知.pdf 片段：046...

参考来源

关于人身保险产品的信息披露渠道，是否有相关的监管政策对保险公司进行规范？

在人身保险产品的

请输入...

中国银保监...

人身保险产...

人身保险产品信息披...

1 / 6

54%



1



2



3



中华人民共和国中央人民政府
www.gov.cn



国家金融监督管理总局规章

来源：银保监会网站

人身保险产品信息披露管理办法

(2022年11月11日中国银行保险监督管理委员会令2022年第8号公布，自2023年6月30日起施行)

第一章 总 则

第一条 为规范人身保险产品信息披露行为，促进行业健康可持续发展，保护投保人、被保险人和受益人的合法权益，根据《中华人民共和国保险法》等法律、行政法规，制定本办法。

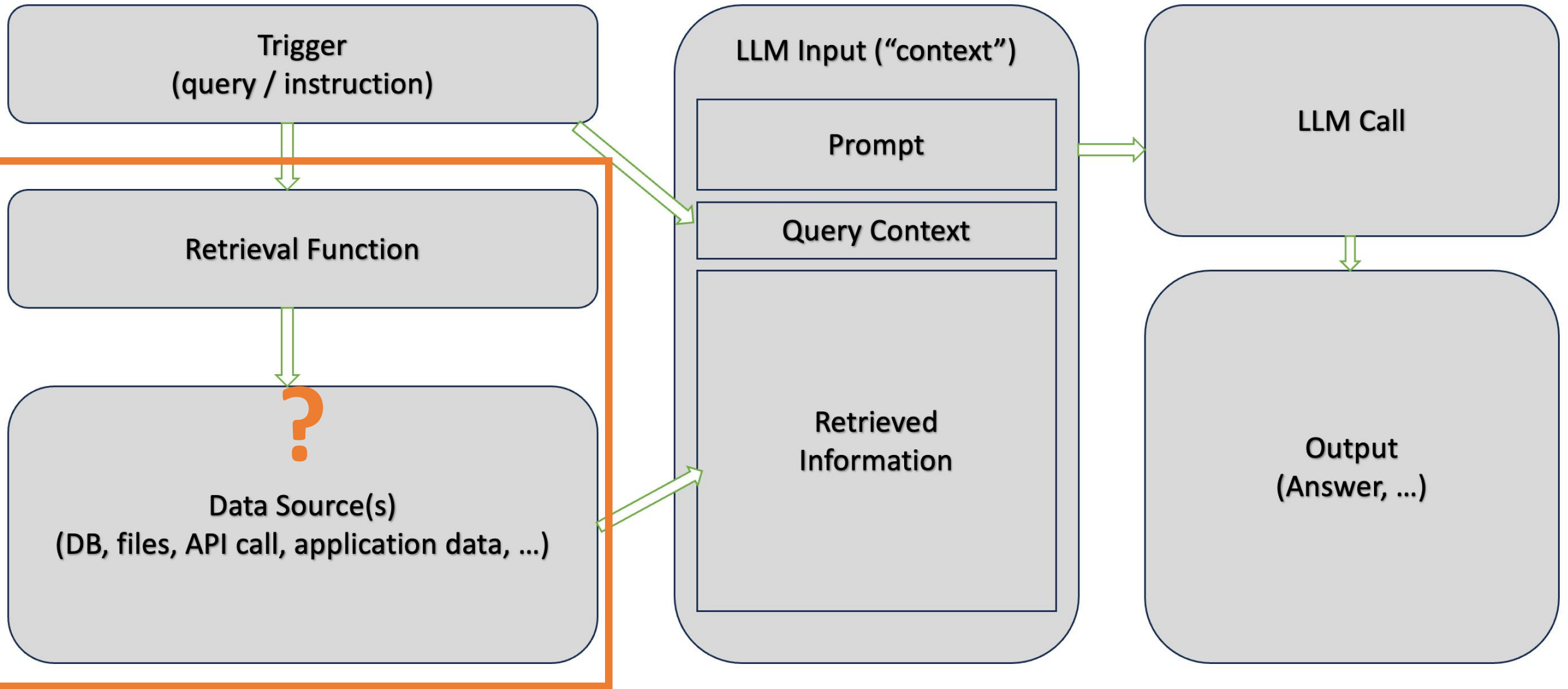
第二条 本办法所称人身保险，按险种类别划分，包括人寿保险、年金保险、健康保险、意外伤害保险等；按设计类型划分，包括普通型、分红型、万能型、投资连结型等。按保险期间划分，包括一年期以上的人身保险和一年期及以下的人身保险。

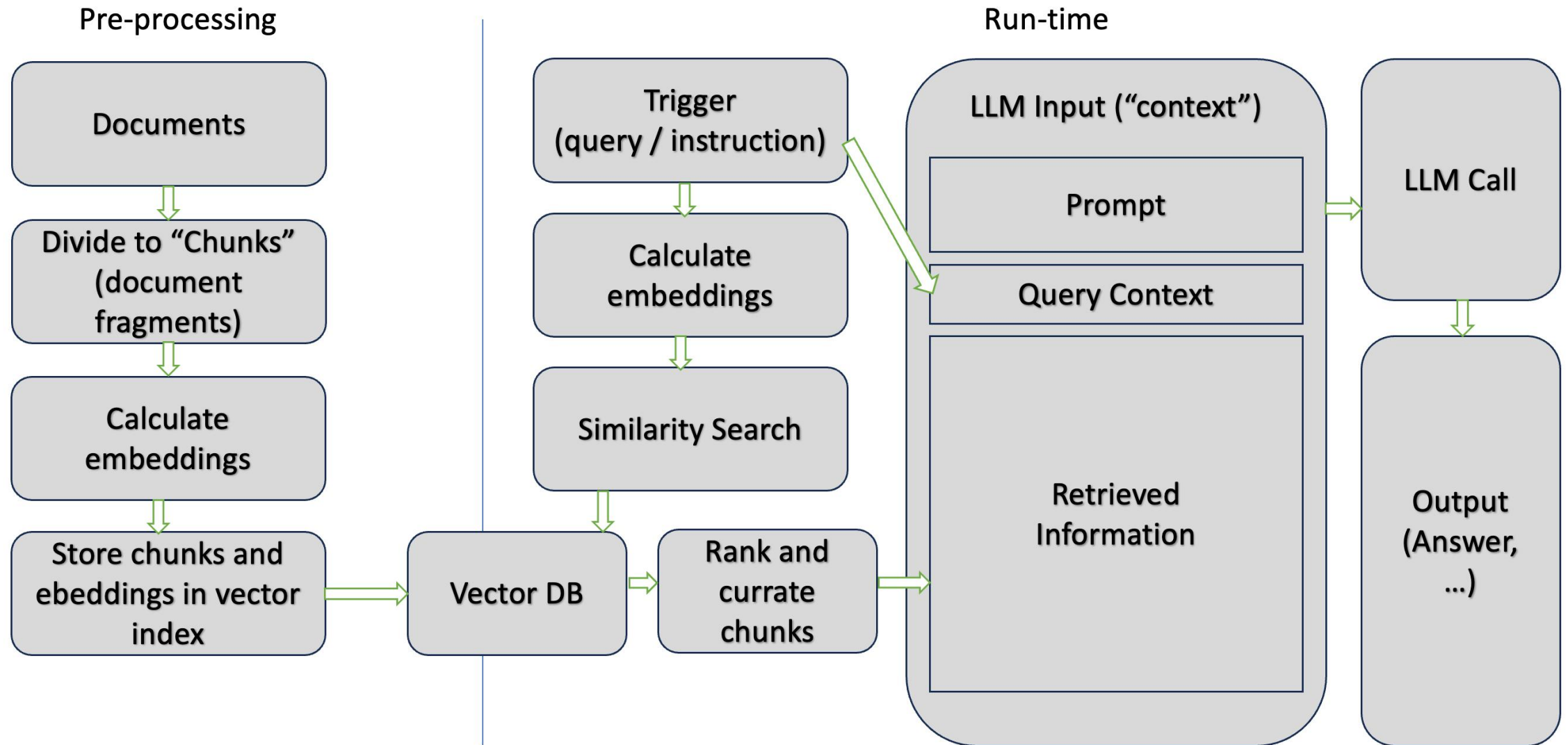
第三条 本办法所称产品信息披露，指保险公司及其保险销售人员、保险中介机构及其从业人员根据法律、行政法规等要求，通过线上或线下等形式，向投保人、被保险人、受益人及社会公众公开保险产品的行为。

第四条 产品信息披露应当遵循真实性、准确性、完整性、及时性原则。保险公司及其保险销售人员、保险中介机构及其从业人员应当准确说明并充分披露与产品相关的信息，无重大遗漏，不得对投保人、被保险人、受益人及社会公众进行隐瞒和欺瞒。

第五条 中国银行保险监督管理委员会（以下简称银保监会）根据法律、行政法规和国务院授权，对保险公司及其保险销售人员、保险中介机构及其从业人员人身保险产品信息披露行为进行监督管理。

第二章 信息披露主体和披露方式





What is Chunking?

- Dividing large documents into smaller segments.
- Chunking is only required if the source documents are too large for the maximum input size imposed by models.
 - For example, the maximum length of input text for the Azure OpenAI text-embedding-ada-002 model is 8,191 tokens.
- Chunks can be defined by document, page, or sentence.

Common Chunking Techniques

- Fixed-size chunks: Define a fixed size that's sufficient for semantically meaningful paragraphs (for example, 200 words) and allows for some overlap (for example, 10-15% of the content) can produce good chunks as input for embedding vector generators.
- Variable-sized chunks based on content: Partition your data based on content characteristics, such as end-of-sentence punctuation marks, end-of-line markers, or using features in the Natural Language Processing (NLP) libraries. Markdown language structure can also be used to split the data.

Importance of Chunk Size in RAG

- Too Large:
 - Loss of Specificity: Generalized embeddings reduce relevance.
 - Information Overload: Excessive data can overwhelm the model.
- Too Small:
 - Loss of Context: Important connections may be lost.
 - Fragmented Information: Leads to incomplete answers.
- Finding the Balance
 - Meaningful & Concise: Retain essential information while fitting processing limits, e.g., append the document title to chunks from the middle of the document to prevent context loss.

Chunk Enrichment

- Cleaning the chunks to support closeness matches by eliminating potential differences that aren't material to the semantics of the text.
- Augmenting the chunks with metadata that can be used as either filters or as part of the search.

Cleaning

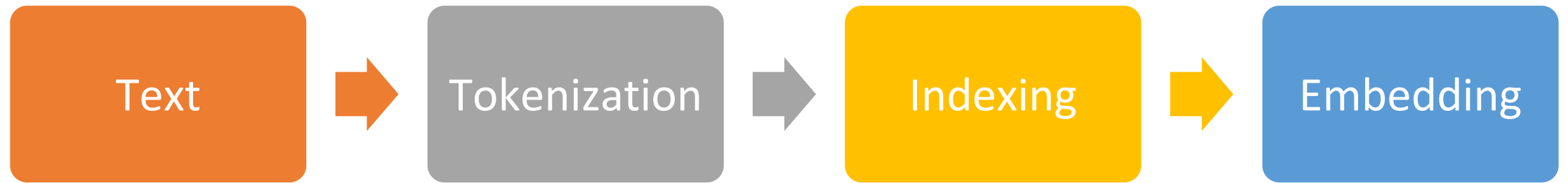
- Lowercasing - Match words regardless of capitalization.
- Remove stop words - Reduce dimensionality by removing common words like "a", "an", "the".
- Fix spelling mistakes - Ensure correct spelling for accurate embeddings.
- Remove Unicode characters - Reduce noise and dimensionality by removing unnecessary characters.
- Normalization - Standardize text by expanding abbreviations and contractions.

Augmenting



Metadata Field	Description	Tools
ID	Uniquely identifies a chunk	Hashing library
Title	Provides a quick content summary, useful for indexed search	Large language model
Summary	Longer than title, useful for indexed search	Large language model
Rephrasing	Captures variations in language for vector search	Large language model
Keywords	Good for exact match searches	Large language model, RAKE, KeyBERT, MultiRake
Entities	Specific info like people, organizations, locations	SpaCy, SNER, scikit-learn, NLTK
Cleaned Text	Clean version of chunk text	Large language model
Answerable Questions	Formulated queries for better matching	Large language model
Source	Original source information	N/A
Language	Filter queries by language	N/A

Text Embedding



Tokens	Characters
6	20

What is a tokenizer?

[2061, 318, 257, 11241, 7509, 30]

Tokenization

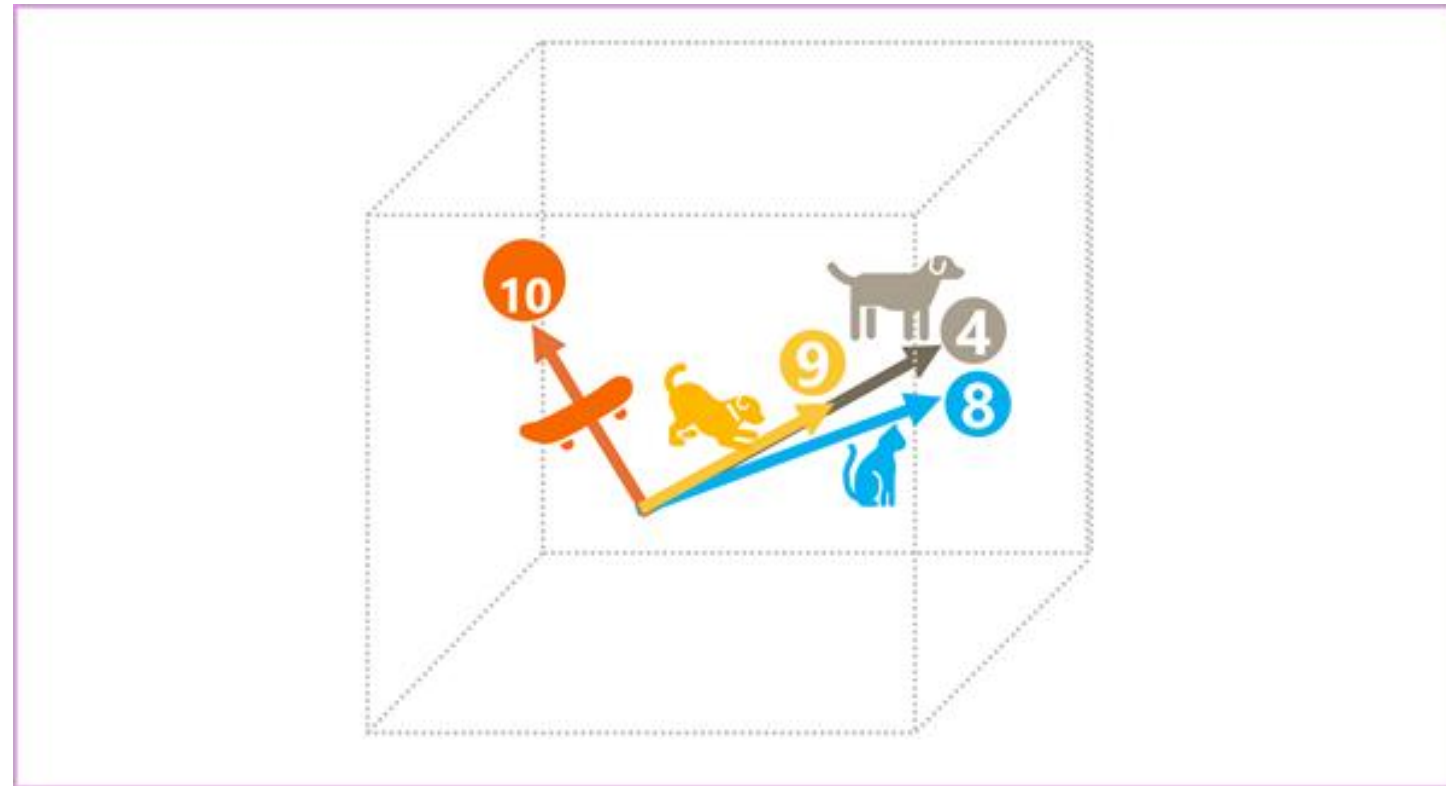
- **Tokens** are common sequences of characters found in a set of text.
- A **tokenizer** is used to split an input text into an array of tokens.
- The exact **tokenization** process varies between models.
- Each token is mapped with a **token index**, which is the integer encoding of the original text chunk.
- One token generally corresponds to ~4 characters of text for common English text. This translates to roughly $\frac{3}{4}$ of a word (so 100 tokens $\approx$ 75 words).

Word Embeddings

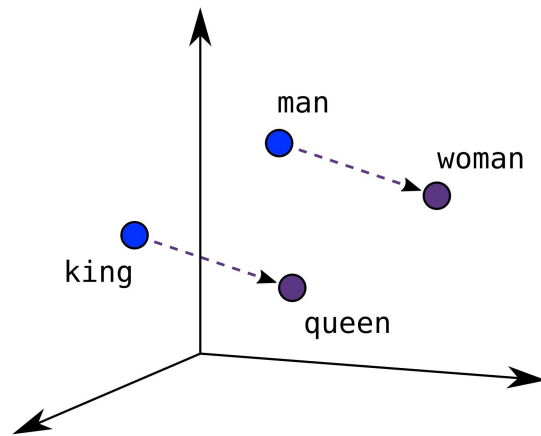
- While it may be convenient to represent tokens as **simple IDs - essentially creating an index for all the words in the vocabulary**, they don't tell us anything about the meaning of the words, or the relationships between them.
- To create a vocabulary that encapsulates **semantic** relationships between the tokens, we define contextual vectors, known as embeddings, for them.
- Vectors are multi-valued numeric representations of information, for example [10, 3, 1] in which each numeric element represents a particular attribute of the information.
- Vectors represent lines in multidimensional space, describing *direction* and *distance* (or amplitude and magnitude) along multiple axes.

Word Embeddings

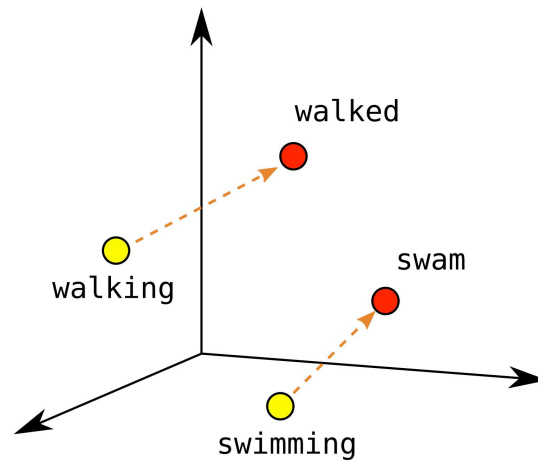
- 4 ("dog"): [10,3,2]
- 8 ("cat"): [10,3,1]
- 9 ("puppy"): [5,2,1]
- 10 ("skateboard"): [-3,3,2]
- Semantically similar tokens point in the same direction.



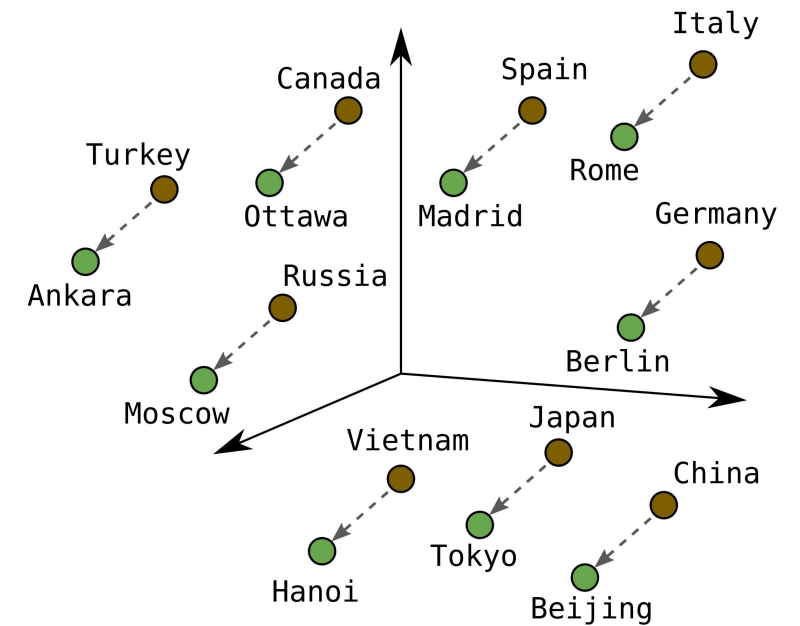
Position (distance and direction) in the vector space can encode semantics.



Male-Female



Verb Tense

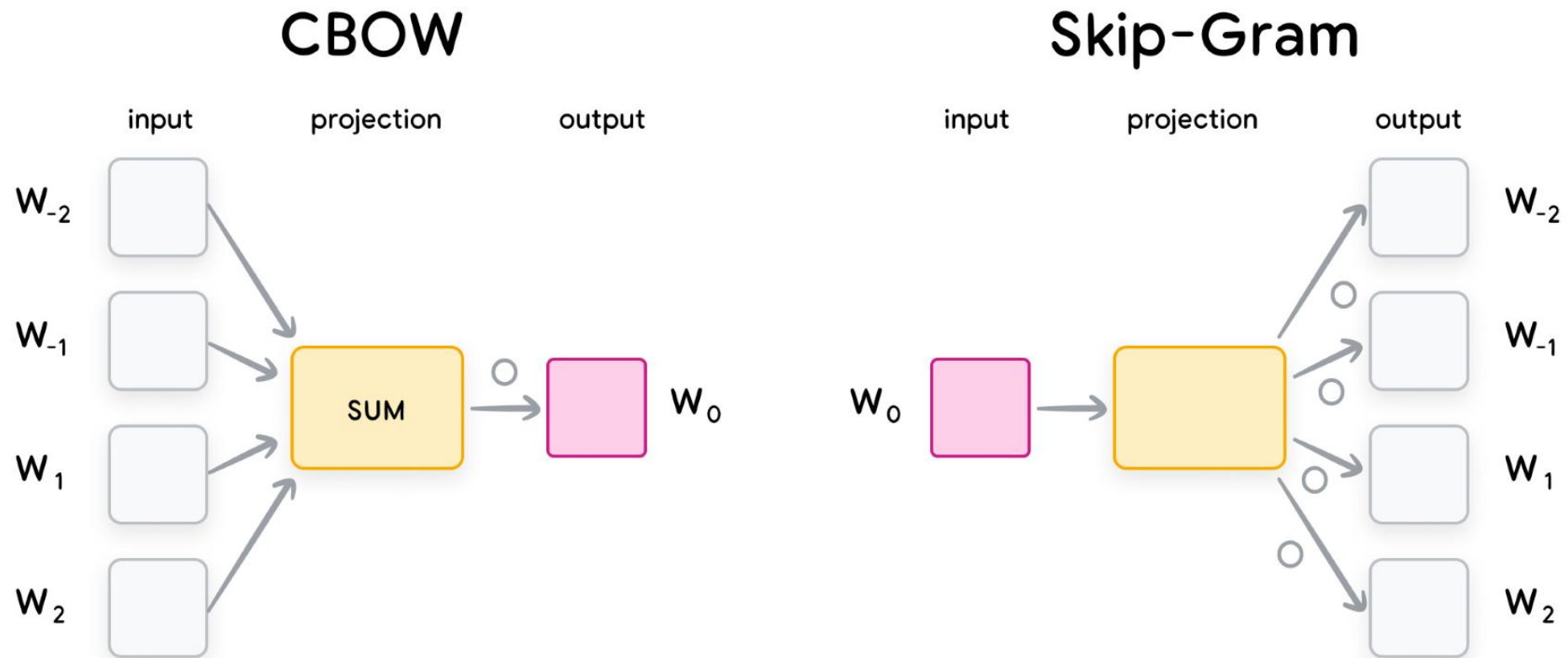


Country-Capital

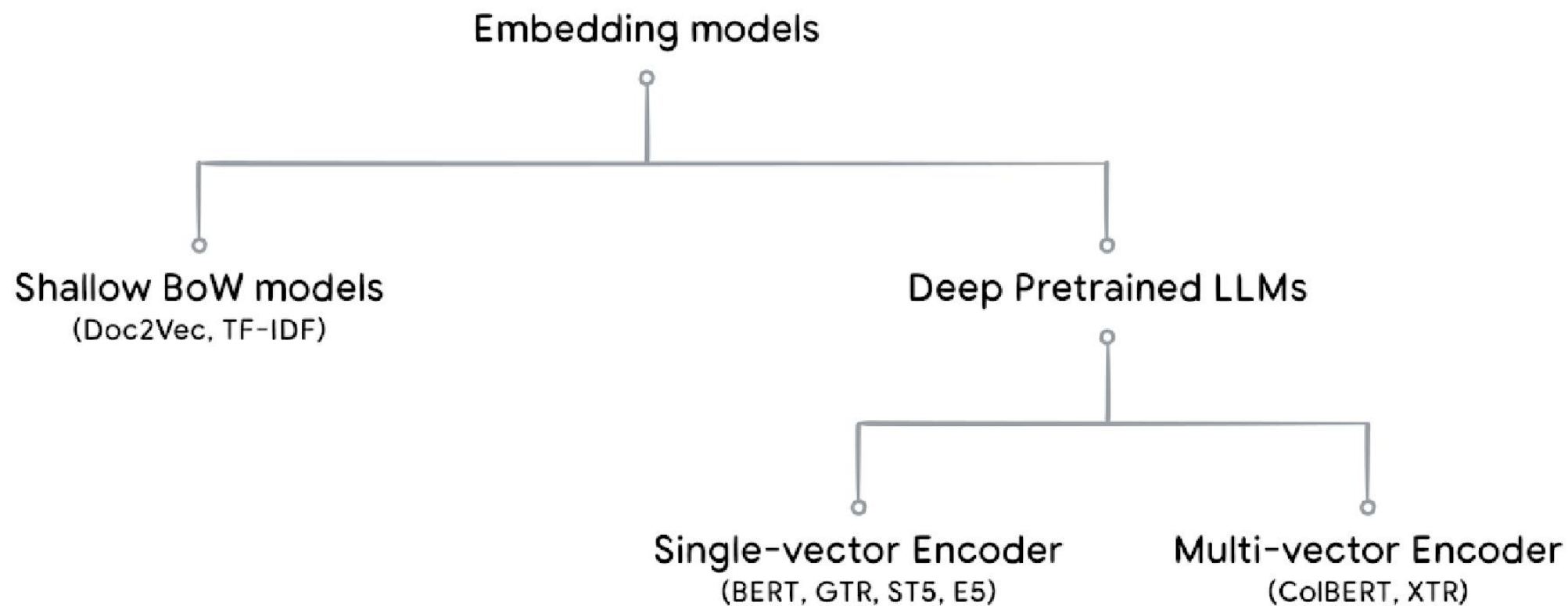
Word Embedding Techniques

- Word2Vec
 - Principle: “the semantic meaning of a word is defined by its neighbors”
 - Two approaches: Continuous bag of words (CBOW) and Skip-Gram
- GloVe (Global Vectors for Word Representation)
 - Captures global statistical information by analyzing word co-occurrence matrices
- FastText
 - Considers subword information (like character n-grams), allowing it to create embeddings for out-of-vocabulary words

Word2Vec's Two Approaches



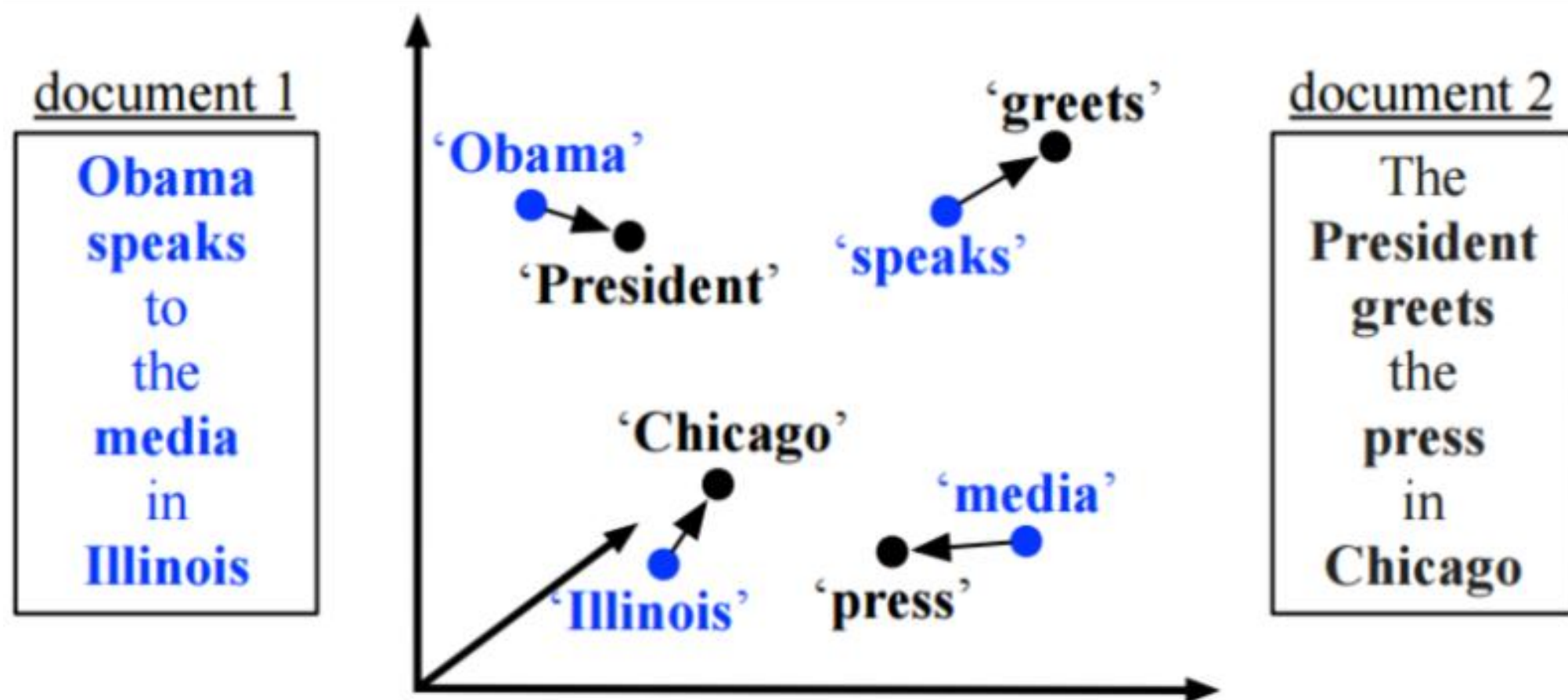
Document Embeddings



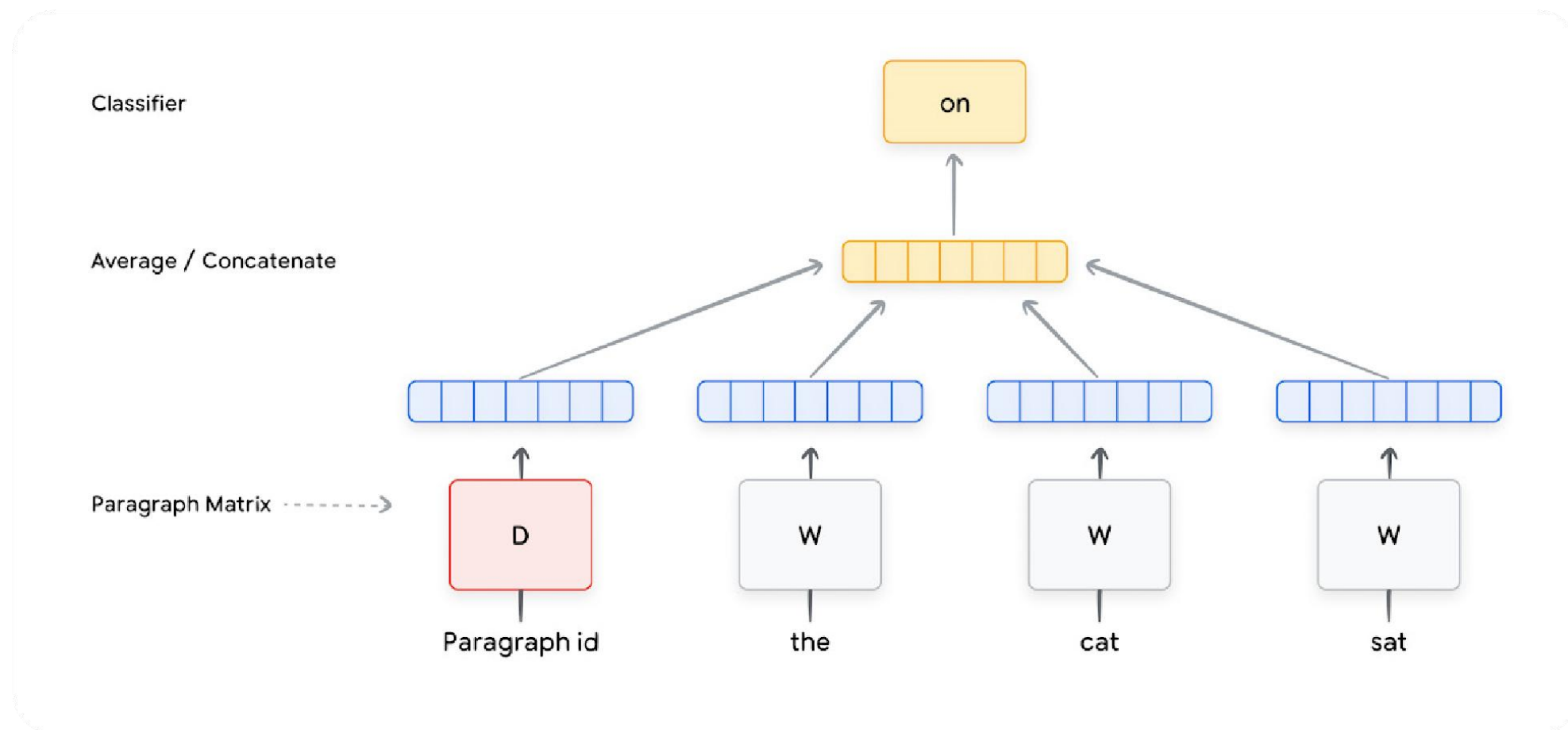
Shallow Bag-of-Words (BoW) models

- Statistical models that use the word frequency to represent the document embedding
- TF-IDF: term frequency-inverse document frequency
 - Adjust for the weights of popular and niche words
- Major weakness: both the word ordering and the semantic meanings are ignored.

Limitations of BoW Models



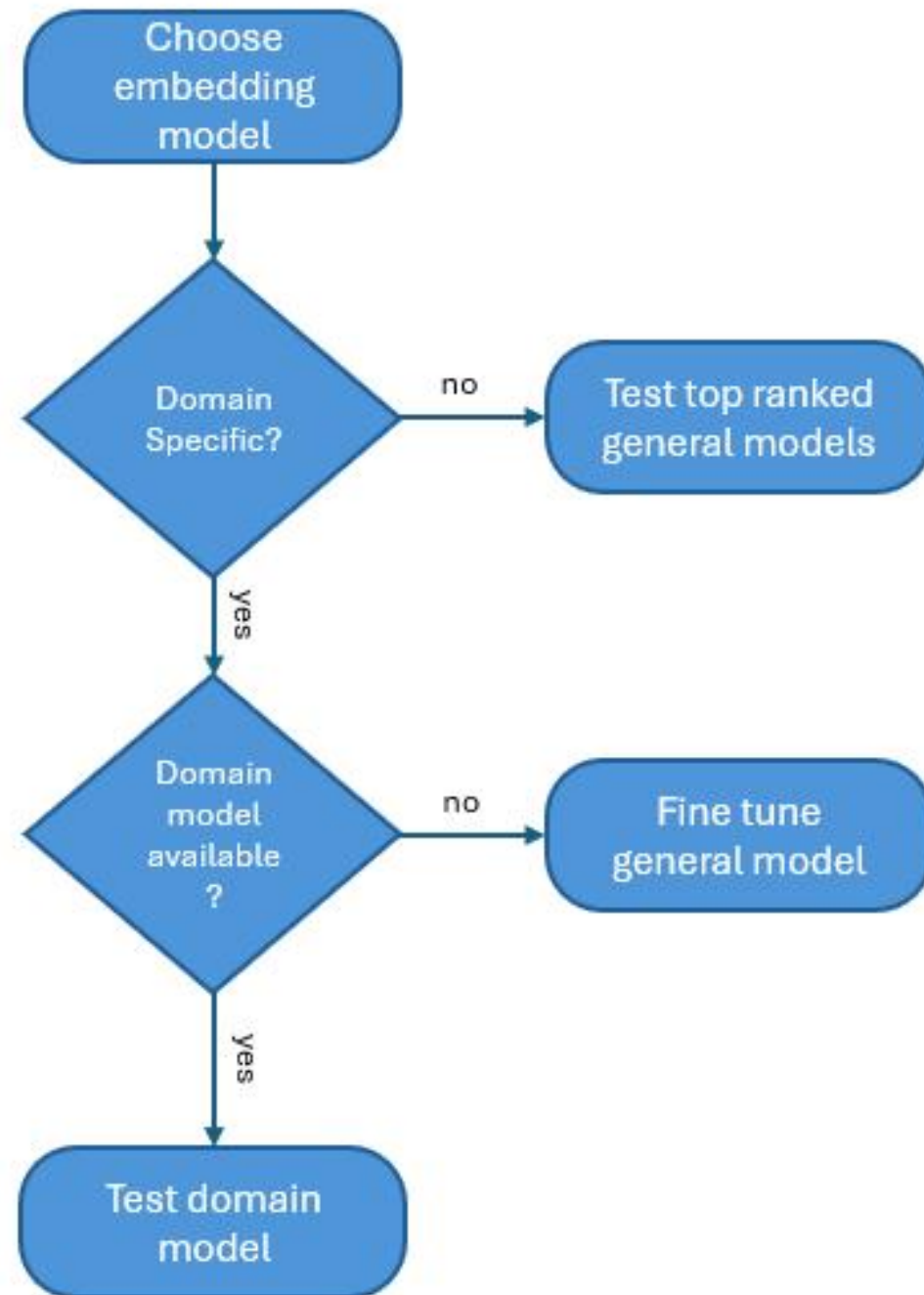
Doc2Vec CBOW model



Deep Pretrained LLMs

- Using more complex learning models, especially bi-directional deep neural network models.
- The use of massive pre-training on unlabeled text.
- The use of a subword tokenizer.
- Using fine-tuning for various downstream NLP tasks.

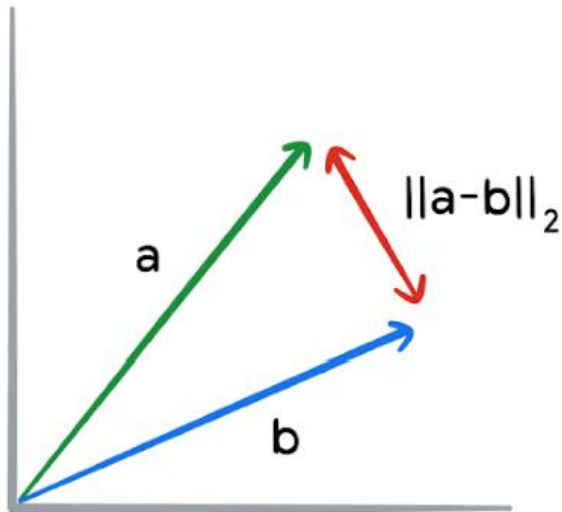
Choosing an embedding model



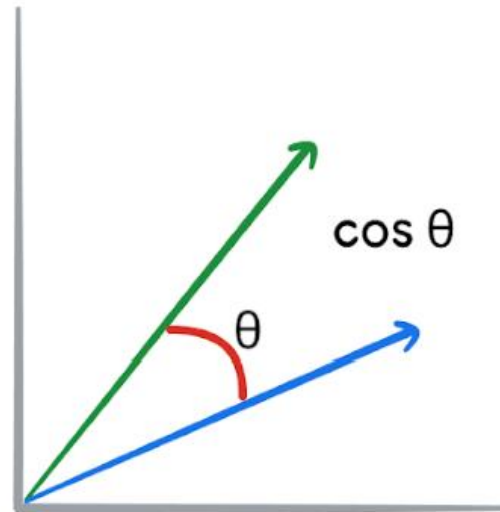
Vector Search

- Full-text Keyword Search Limitations
 - Relies on explicit keyword matching
 - Fails with misspellings or different wording
- Advantages of Vector Search
 - Searches for meaning across data modalities
 - Provides nuanced results

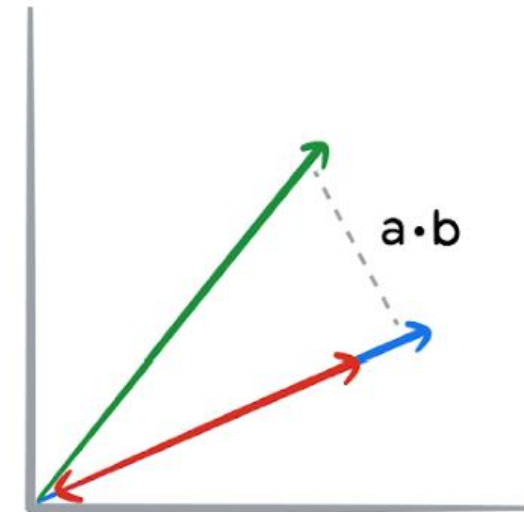
How different metrics compute vector similarity



L2 distance



Cosine similarity



Inner product

Vector Similarity

- Euclidean distance (i.e., L2 distance) is a geometric measure of the distance between two points in a vector space.
- Cosine similarity is a measure of the angle between two vectors
- Inner/dot product, is the projection of one vector onto another.

Similar sentences

The angle between the two embedding vectors is small.

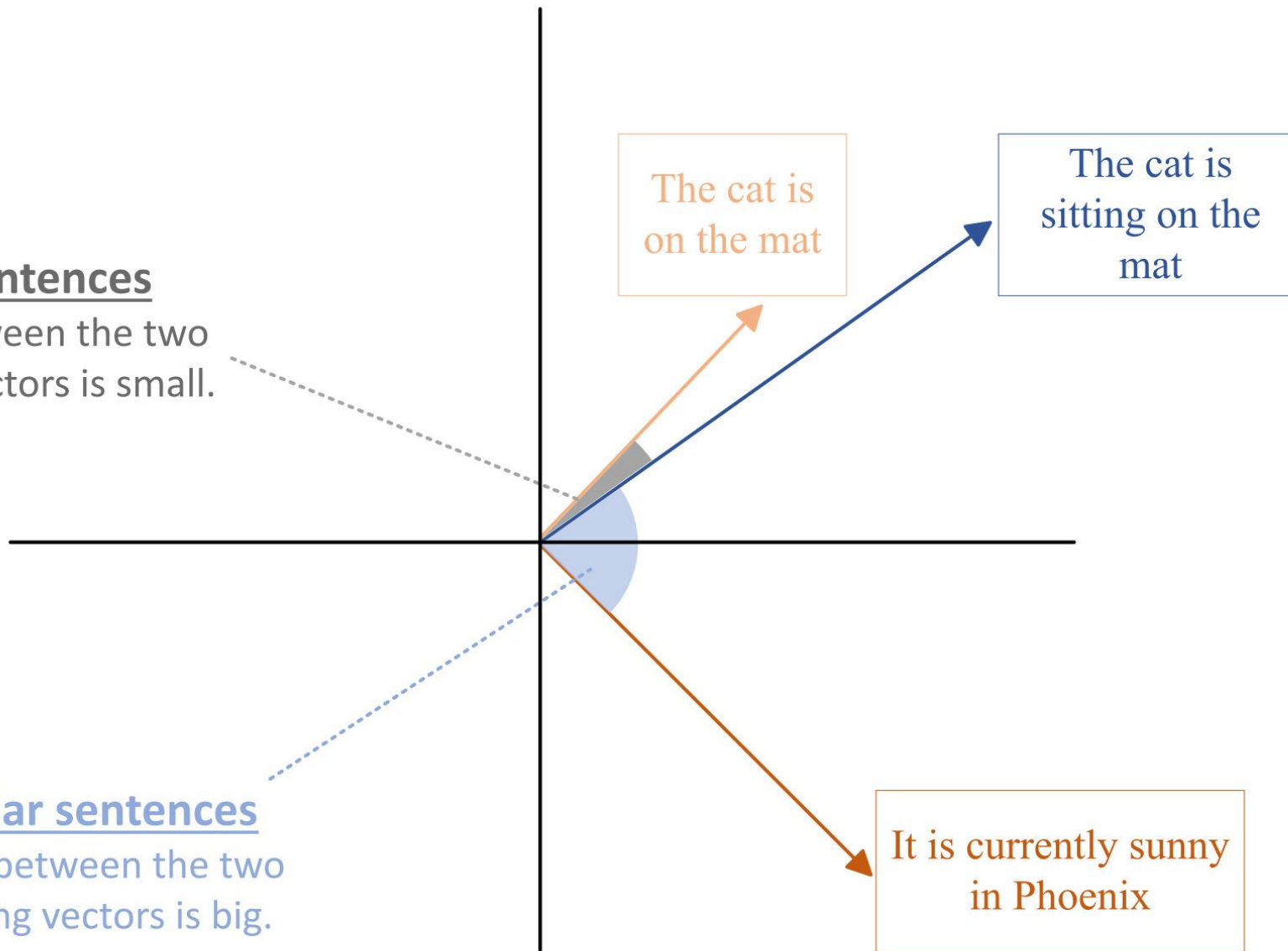
The cat is
on the mat

The cat is
sitting on the
mat

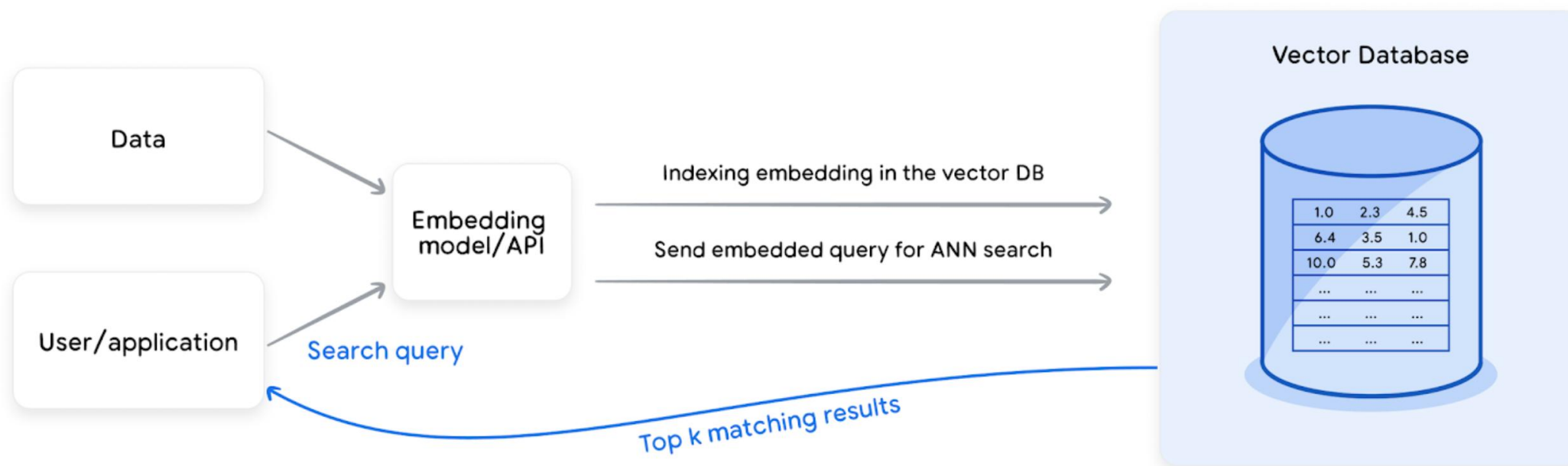
Dissimilar sentences

The angle between the two embedding vectors is big.

It is currently sunny
in Phoenix



Populating and Querying Vector Databases



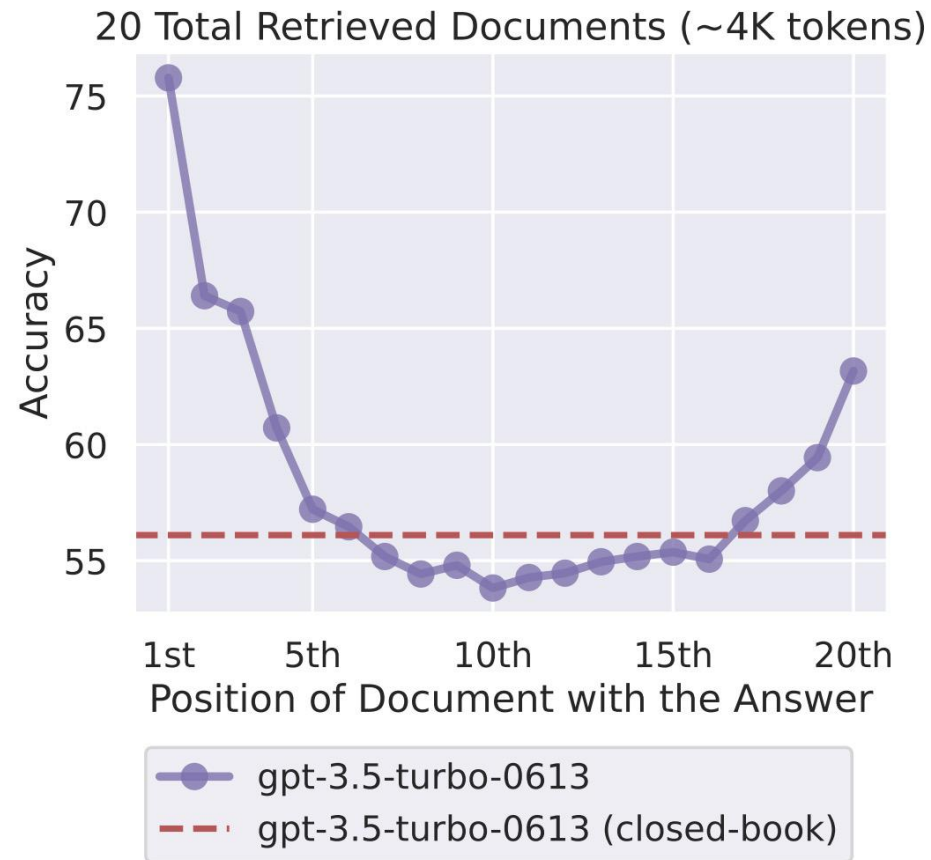
Populating and Querying Vector Databases

- Embedding: Use a trained embedding model to convert data points into fixed-dimension vectors.
- Indexing: Augment vectors with metadata and index them for efficient search.
- Querying: Embed incoming queries and retrieve the most semantically similar items. Utilize caching, pre-filtering, and post-filtering to enhance performance.

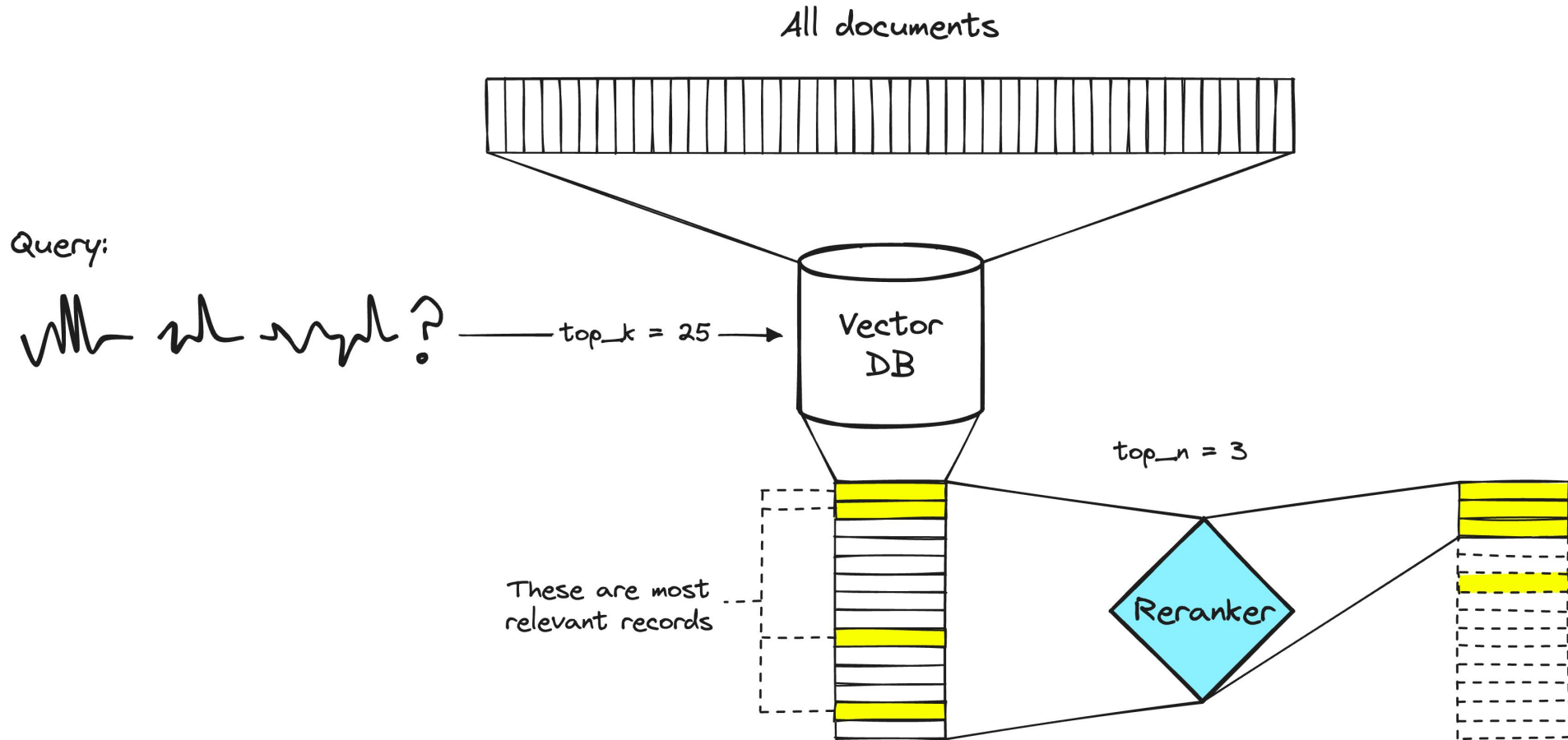
Vector Search Algorithms

- Linear Search
 - Compares query vector with each document vector
 - Returns the one with the highest similarity
 - Scales linearly ($O(N)$) with the number of documents
 - Unacceptably slow for large datasets
- Approximate Nearest Neighbour (ANN) Search
 - Finds closest points with a small margin of error
 - Boosts performance significantly
 - Uses techniques like quantization, hashing, clustering, and trees
 - Offers varying trade-offs across scale, indexing time, performance, and simplicity

Context Stuffing reduces the LLM's recall performance



A two-stage retrieval system



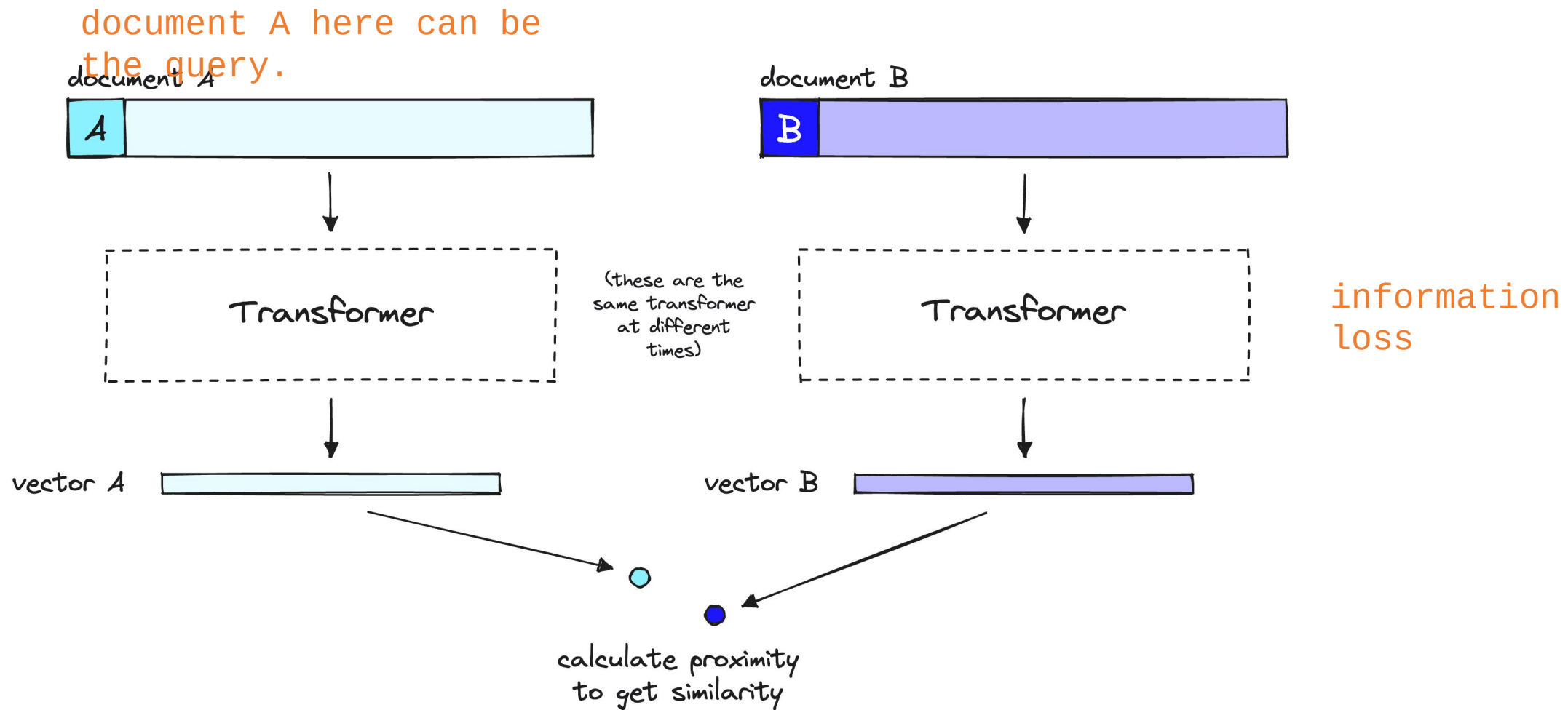
A two-stage retrieval system

- A first-stage model (an embedding model/retriever) retrieves a set of relevant documents from a larger dataset.
- A second-stage model (the reranker) is used to rerank those documents retrieved by the first-stage model.

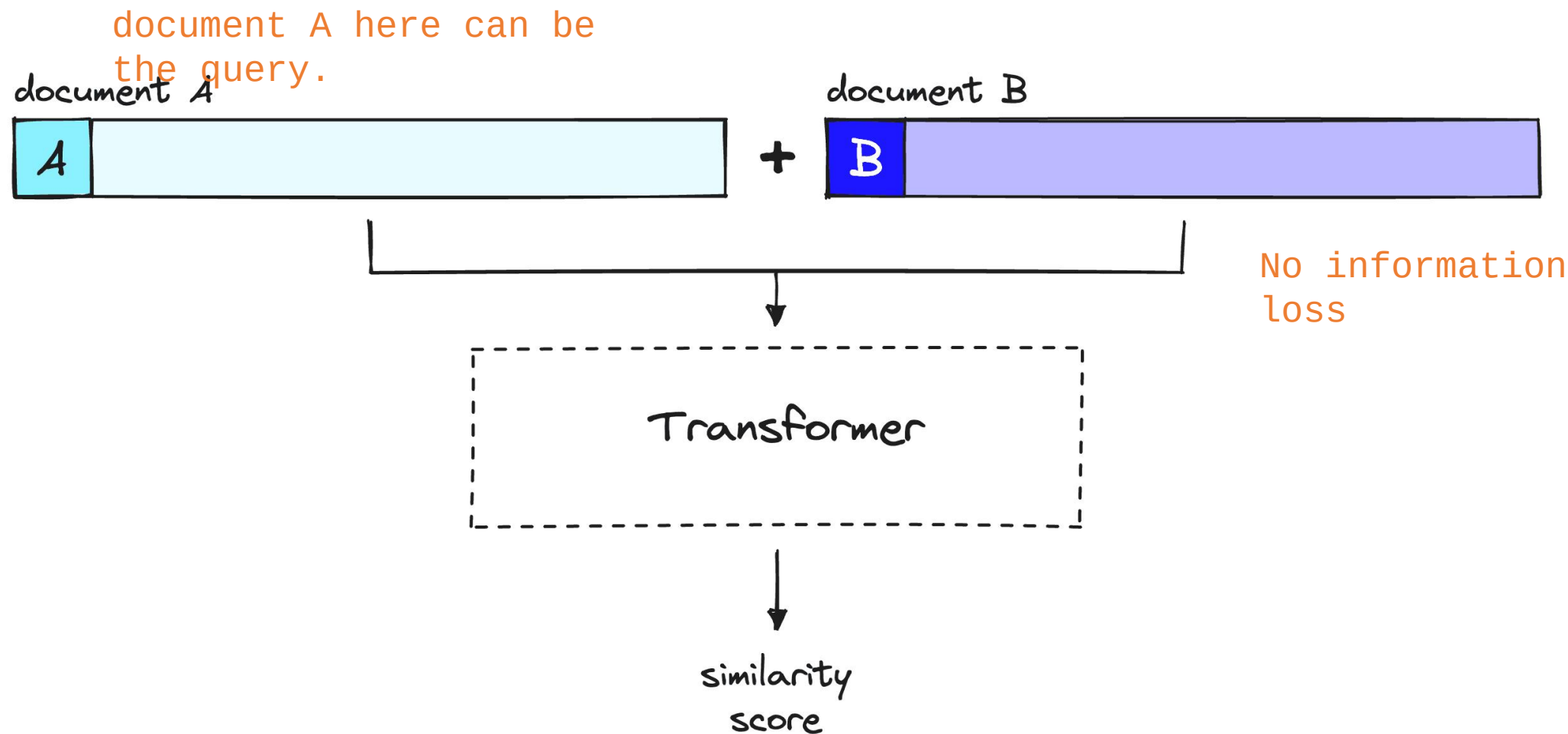
Reranking

- A reranking model — also known as a cross-encoder — is a type of model that, given a query and document pair, will output a similarity score.
- We use this score to reorder the documents by relevance to our query.

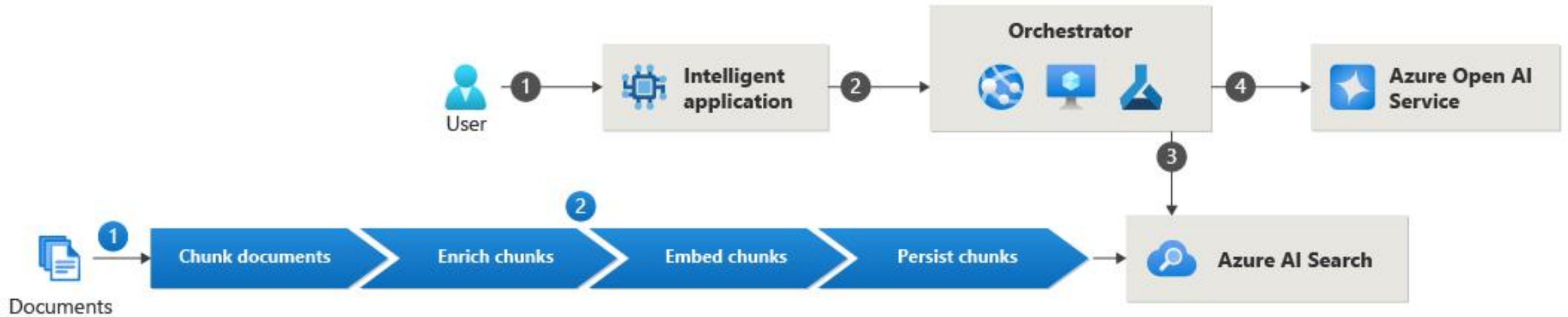
A bi-encoder model

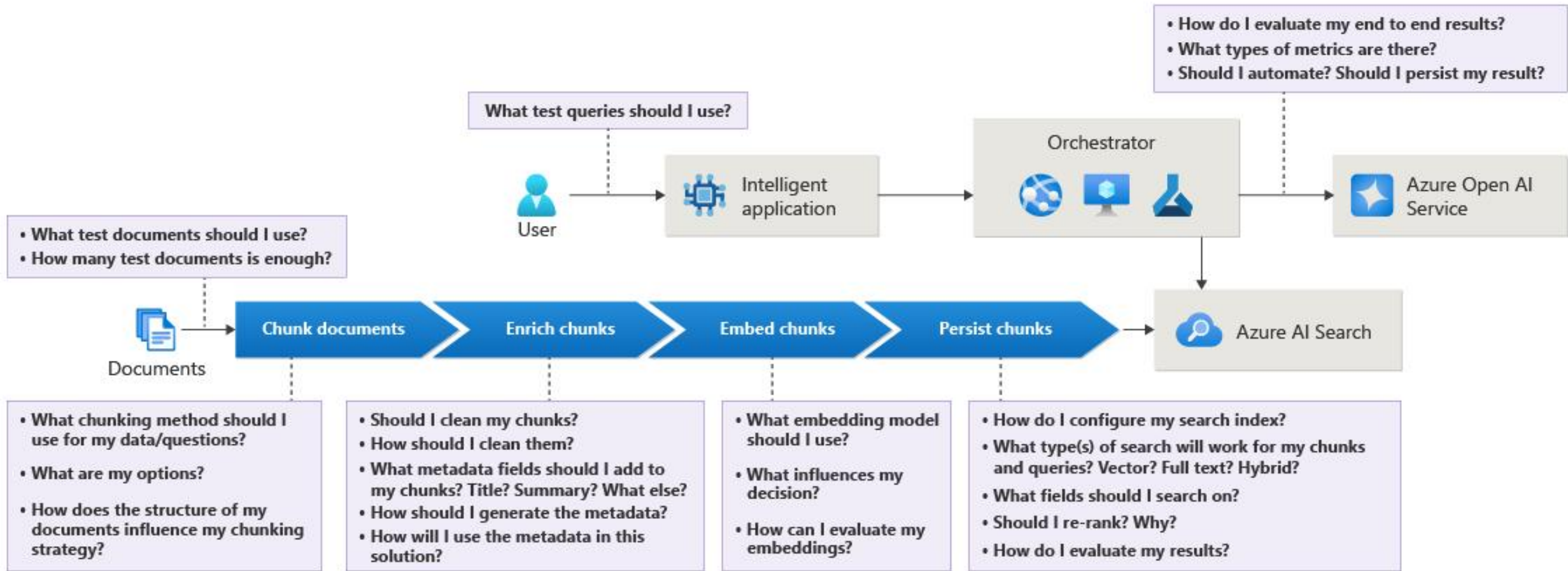


A cross-encoder model (reranker)



Azure RAG Architecture



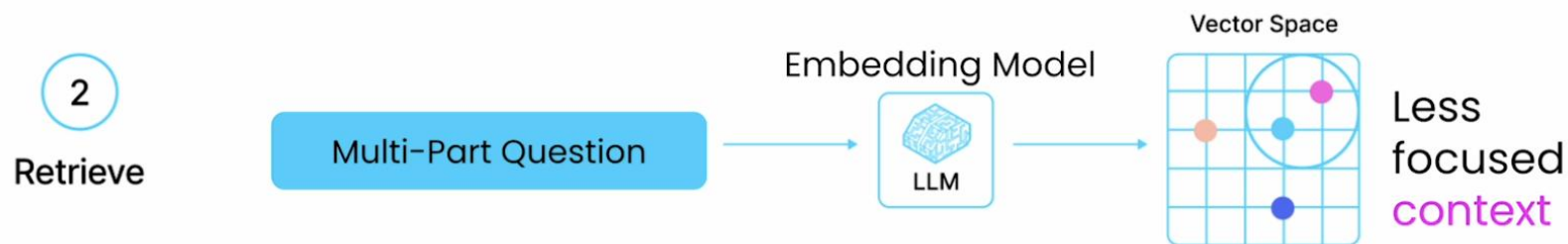


RAG Evaluation

Measure	Description
Groundedness	Measures whether the response is completely based on the context
Completeness	Measures whether the response is answering all parts of the query
Utilization	Measures the extent to which the response is made up of information from the chunks in the context
Relevance	Measures the extent to which the response is pertinent and related to the query

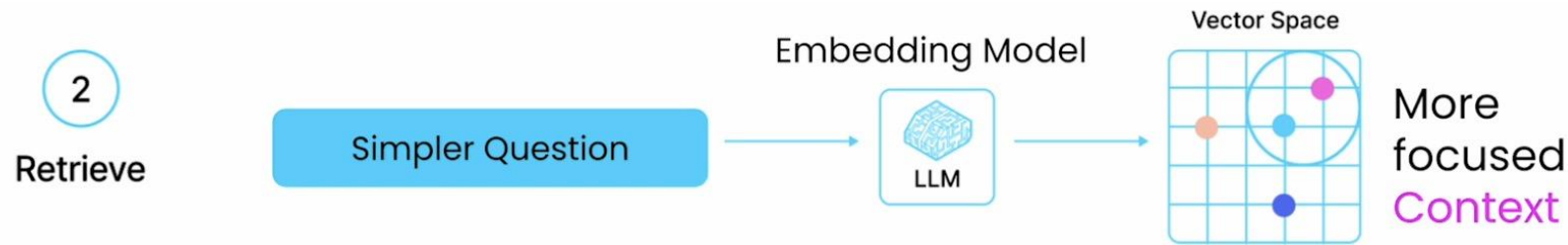
Limitations of RAG

- Summarization
- Comparison
- Multi-part questions



Splitting up complex questions

- Split up multi-part questions into simpler questions



- Task for LLM:
 - Split up a complicated question
 - Glue the answers back together

Building an agent